

Chapter 11 — Runtime, Resources, Operations, Failure, Recovery, and Change

Purpose and scope

Defines runtime state, health, readiness, supervision, resources, degradation, incidents, recovery objectives, releases, migrations, and exceptions.

Source authority

Book I: Book I continuity, reversibility, restraint, accountability, and constitutional shutdown

Book II: Book II Chapters 02, 22, 27, 28, 29, 33, 34, and 35

Book III: Book III Chapters 01, 02, 04, 06, 07, 08, and 09

Normative semantic rules

- Runtime health MUST distinguish Liveness, Readiness, Health, Desired State, and Observed State.
- Degraded, Safe, Restricted, Quarantined, and Recovering modes MUST be explicit, observable, and governed by transition criteria.
- Recovery MUST restore identity, authority, state, evidence, and trust as applicable; restart alone MUST NOT be called Recovery.
- Exceptions MUST be time-bounded and MUST NOT waive Constitutional Invariants; Architecture Deviations MUST use architecture governance.

Canonical terms

HAL-TERM-0130 — Runtime

The governed combination of processes, state, resources, policies, dependencies, and environments in which HAL behavior executes.

Distinction: A Runtime is not HAL's constitutional identity.

Type/category/status: Execution environment / Operations / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51

Book II: Book II Chapters 02, 27, 28, 33, and 34

Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0130` when it uses **Runtime** with this exact governed meaning: The governed combination of processes, state, resources, policies, dependencies, and environments in which HAL behavior executes.

Counterexample: A dependent artifact uses **Runtime** in a way that violates its required distinction: A Runtime is not HAL's constitutional identity.

Relationships: None registered

Lifecycle: None registered

Constraint: A Runtime is not HAL's constitutional identity.

HAL-TERM-0131 — Node

A governed compute or device participant capable of hosting workload, state, sensing, or action under the runtime architecture.

Distinction: A Node does not independently become HAL or own authoritative state without explicit designation.

Type/category/status: Execution entity / Operations / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51

Book II: Book II Chapters 02, 27, 28, 33, and 34

Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0131` when it uses **Node** with this exact governed meaning: A governed compute or device participant capable of hosting workload, state, sensing, or action under the runtime

architecture.

Counterexample: A dependent artifact uses ****Node**** in a way that violates its required distinction: A Node does not independently become HAL or own authoritative state without explicit designation.

Relationships: None registered

Lifecycle: None registered

Constraint: A Node does not independently become HAL or own authoritative state without explicit designation.

HAL-TERM-0132 — Service

A deployable runtime boundary providing one or more governed responsibilities or interfaces.

Distinction: A Service is not automatically an architectural component, capability, or authority domain.

Type/category/status: Component deployment role / Operations / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51

Book II: Book II Chapters 02, 27, 28, 33, and 34

Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0132` when it uses ****Service**** with this exact governed meaning: A deployable runtime boundary providing one or more governed responsibilities or interfaces.

Counterexample: A dependent artifact uses ****Service**** in a way that violates its required distinction: A Service is not automatically an architectural component, capability, or authority domain.

Relationships: HAL-REL-0044

Lifecycle: HAL-TRANS-0021, HAL-TRANS-0022, HAL-TRANS-0023, HAL-TRANS-0024

Constraint: A Service is not automatically an architectural component, capability, or authority domain.

HAL-TERM-0133 — Supervisor

A component that monitors and governs workload lifecycle, desired state, health, restart, containment, and escalation within its authority.

Distinction: A Supervisor must not conceal repeated failure through unlimited restart loops.

Type/category/status: Runtime control role / Operations / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51

Book II: Book II Chapters 02, 27, 28, 33, and 34

Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0133` when it uses ****Supervisor**** with this exact governed meaning: A component that monitors and governs workload lifecycle, desired state, health, restart, containment, and escalation within its authority.

Counterexample: A dependent artifact uses ****Supervisor**** in a way that violates its required distinction: A Supervisor must not conceal repeated failure through unlimited restart loops.

Relationships: HAL-REL-0044

Lifecycle: None registered

Constraint: A Supervisor must not conceal repeated failure through unlimited restart loops.

HAL-TERM-0134 — Desired State

The governed runtime condition an authorized controller intends the system to maintain.

Distinction: Desired State is not proof of Observed State or successful convergence.

Type/category/status: Control target / Operations / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51

Book II: Book II Chapters 02, 27, 28, 33, and 34

Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0134` when it uses ****Desired State**** with this exact governed meaning: The governed runtime condition an authorized controller intends the system to maintain.

Counterexample: A dependent artifact uses ****Desired State**** in a way that violates its required distinction: Desired State is not proof of Observed State or successful convergence.

Relationships: HAL-REL-0046

Lifecycle: None registered

Constraint: Desired State is not proof of Observed State or successful convergence.

HAL-TERM-0135 — Observed State

The evidenced runtime condition measured at a defined time and scope.

Distinction: Observation may be stale, partial, or uncertain and must not be treated as Desired State.

Type/category/status: Runtime observation / Operations / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51

Book II: Book II Chapters 02, 27, 28, 33, and 34

Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0135` when it uses ****Observed State**** with this exact governed meaning: The evidenced runtime condition measured at a defined time and scope.

Counterexample: A dependent artifact uses ****Observed State**** in a way that violates its required distinction: Observation may be stale, partial, or uncertain and must not be treated as Desired State.

Relationships: None registered

Lifecycle: None registered

Constraint: Observation may be stale, partial, or uncertain and must not be treated as Desired State.

HAL-TERM-0136 — Health

A multidimensional assessment of a component's ability to perform its declared responsibilities within current constraints.

Distinction: Process liveness alone is not Health.

Type/category/status: Operational assessment / Operations / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51

Book II: Book II Chapters 02, 27, 28, 33, and 34

Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0136` when it uses ****Health**** with this exact governed meaning: A multidimensional assessment of a component's ability to perform its declared responsibilities within current constraints.

Counterexample: A dependent artifact uses ****Health**** in a way that violates its required distinction: Process liveness alone is not Health.

Relationships: None registered

Lifecycle: None registered

Constraint: Process liveness alone is not Health.

HAL-TERM-0137 — Liveness

Evidence that a workload is running or able to make progress according to a narrow declared probe.

Distinction: Liveness does not establish readiness, correctness, authorization, or safety.

Type/category/status: Runtime signal / Operations / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51

Book II: Book II Chapters 02, 27, 28, 33, and 34

Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0137` when it uses ****Liveness**** with this exact governed meaning: Evidence that a workload is running or able to make progress according to a narrow declared probe.

Counterexample: A dependent artifact uses ****Liveness**** in a way that violates its required distinction: Liveness does not establish readiness, correctness, authorization, or safety.

Relationships: None registered

Lifecycle: None registered

Constraint: Liveness does not establish readiness, correctness, authorization, or safety.

HAL-TERM-0138 — Readiness

Evidence that a workload may receive its declared class of work under current dependencies, configuration, and safety conditions.

Distinction: Readiness is scoped and must not be inferred from Liveness.

Type/category/status: Admission signal / Operations / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51

Book II: Book II Chapters 02, 27, 28, 33, and 34

Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0138` when it uses ****Readiness**** with this exact governed meaning: Evidence that a workload may receive its declared class of work under current dependencies, configuration, and safety

conditions.

Counterexample: A dependent artifact uses ****Readiness**** in a way that violates its required distinction: Readiness is scoped and must not be inferred from Liveness.

Relationships: None registered

Lifecycle: None registered

Constraint: Readiness is scoped and must not be inferred from Liveness.

HAL-TERM-0139 — Resource

A bounded consumable or allocatable asset such as compute, memory, storage, bandwidth, attention, time, energy, or device capacity.

Distinction: Resource availability does not imply authority to allocate or consume it.

Type/category/status: Governed capacity / Operations / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51

Book II: Book II Chapters 02, 27, 28, 33, and 34

Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0139` when it uses ****Resource**** with this exact governed meaning: A bounded consumable or allocatable asset such as compute, memory, storage, bandwidth, attention, time, energy, or device capacity.

Counterexample: A dependent artifact uses ****Resource**** in a way that violates its required distinction: Resource availability does not imply authority to allocate or consume it.

Relationships: HAL-REL-0045

Lifecycle: None registered

Constraint: Resource availability does not imply authority to allocate or consume it.

HAL-TERM-0140 — Reservation

A time-bounded governed allocation claim against a Resource for a declared purpose and owner.

Distinction: A Reservation is not proof the resource was consumed or the work completed.

Type/category/status: Resource claim / Operations / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51

Book II: Book II Chapters 02, 27, 28, 33, and 34

Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0140` when it uses ****Reservation**** with this exact governed meaning: A time-bounded governed allocation claim against a Resource for a declared purpose and owner.

Counterexample: A dependent artifact uses ****Reservation**** in a way that violates its required distinction: A Reservation is not proof the resource was consumed or the work completed.

Relationships: HAL-REL-0045

Lifecycle: None registered

Constraint: A Reservation is not proof the resource was consumed or the work completed.

HAL-TERM-0141 — Degraded Mode

A declared operating state in which selected capabilities or service levels are reduced while higher-priority constitutional, authority, safety, privacy, and evidence obligations remain protected.

Distinction: Degradation must not silently weaken non-degradable controls.

Type/category/status: Operating state / Operations / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51

Book II: Book II Chapters 02, 27, 28, 33, and 34

Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0141` when it uses ****Degraded Mode**** with this exact governed meaning: A declared operating state in which selected capabilities or service levels are reduced while higher-priority constitutional, authority, safety, privacy, and evidence obligations remain protected.

Counterexample: A dependent artifact uses ****Degraded Mode**** in a way that violates its required distinction: Degradation must not silently weaken non-degradable controls.

Relationships: None registered

Lifecycle: None registered

Constraint: Degradation must not silently weaken non-degradable controls.

HAL-TERM-0142 — Safe Mode

A constrained operating state that prioritizes containment, Owner communication, essential evidence, and prevention of unauthorized or unsafe effects.

Distinction: Safe Mode is not a generic low-performance mode.

Type/category/status: Protective operating state / Operations / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51

Book II: Book II Chapters 02, 27, 28, 33, and 34

Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0142` when it uses ****Safe Mode**** with this exact governed meaning: A constrained operating state that prioritizes containment, Owner communication, essential evidence, and prevention of unauthorized or unsafe effects.

Counterexample: A dependent artifact uses ****Safe Mode**** in a way that violates its required distinction: Safe Mode is not a generic low-performance mode.

Relationships: None registered

Lifecycle: None registered

Constraint: Safe Mode is not a generic low-performance mode.

HAL-TERM-0143 — Restricted Mode

An operating state in which selected capabilities, integrations, or authority paths are disabled or narrowed because required trust, evidence, policy, or assurance is unavailable.

Distinction: It must be explicit, observable, and reversible through governed recovery.

Type/category/status: Authority-limited state / Operations / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 29, 38, 42, 47, and 51

Book II: Book II Chapters 02, 27, 28, 33, and 34

Book III: Book III Chapters 3, 4, 6, and 7

Example: A dependent artifact cites `HAL-TERM-0143` when it uses ****Restricted Mode**** with this exact governed meaning: An operating state in which selected capabilities, integrations, or authority paths are disabled or narrowed because required trust, evidence, policy, or assurance is unavailable.

Counterexample: A dependent artifact uses ****Restricted Mode**** in a way that violates its required distinction: It must be explicit, observable, and reversible through governed recovery.

Relationships: None registered

Lifecycle: None registered

Constraint: It must be explicit, observable, and reversible through governed recovery.

HAL-TERM-0144 — Incident

An event or condition requiring coordinated response because it threatens constitutional conformance, authority, security, privacy, trust, availability, integrity, or material outcomes.

Distinction: An anomaly is not necessarily an Incident until classification criteria are met.

Type/category/status: Governed adverse event / Operations / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 38, 42, and 50

Book II: Book II Chapters 27 and 28

Book III: Book III Chapters 5, 6, 7, and 8

Example: A dependent artifact cites `HAL-TERM-0144` when it uses ****Incident**** with this exact governed meaning: An event or condition requiring coordinated response because it threatens constitutional conformance, authority, security, privacy, trust, availability, integrity, or material outcomes.

Counterexample: A dependent artifact uses ****Incident**** in a way that violates its required distinction: An anomaly is not necessarily an Incident until classification criteria are met.

Relationships: None registered

Lifecycle: None registered

Constraint: An anomaly is not necessarily an Incident until classification criteria are met.

HAL-TERM-0145 — Quarantine

A governed isolation state preventing a component, artifact, identity, message, or data set from participating beyond an explicitly limited inspection boundary.

Distinction: Quarantine is not deletion and must preserve required evidence.

Type/category/status: Containment state / Operations / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 38, 42, and 50

Book II: Book II Chapters 27 and 28

Book III: Book III Chapters 5, 6, 7, and 8

Example: A dependent artifact cites `HAL-TERM-0145` when it uses ****Quarantine**** with this exact governed meaning: A governed isolation state preventing a component, artifact, identity, message, or data set from participating beyond an explicitly limited inspection boundary.

Counterexample: A dependent artifact uses ****Quarantine**** in a way that violates its required distinction: Quarantine is not deletion and must preserve required evidence.

Relationships: None registered

Lifecycle: None registered

Constraint: Quarantine is not deletion and must preserve required evidence.

HAL-TERM-0146 — Recovery

The governed process of restoring acceptable identity, authority, state, service, evidence, and trust after failure or compromise.

Distinction: Restart alone is not Recovery.

Type/category/status: Restoration process / Operations / Approved

Aliases: None

Source basis: Direct source normalization

Book I: Book I Decisions 38, 42, and 50

Book II: Book II Chapters 27 and 28

Book III: Book III Chapters 5, 6, 7, and 8

Example: A dependent artifact cites `HAL-TERM-0146` when it uses ****Recovery**** with this exact governed meaning: The governed process of restoring acceptable identity, authority, state, service, evidence, and trust after failure or compromise.

Counterexample: A dependent artifact uses ****Recovery**** in a way that violates its required distinction: Restart alone is not Recovery.

Relationships: HAL-REL-0046

Lifecycle: None registered

Constraint: Restart alone is not Recovery.

HAL-TERM-0147 — Recovery Point Objective

The maximum tolerable loss or unavailability of recoverable state measured from the disruption point for a declared domain.

Distinction: RPO is a target, not proof that recovery met it.

Type/category/status: Recovery target / Operations / Approved

Aliases: RPO

Source basis: Direct source normalization

Book I: Book I Decisions 38, 42, and 50

Book II: Book II Chapters 27 and 28

Book III: Book III Chapters 5, 6, 7, and 8

Example: A dependent artifact cites `HAL-TERM-0147` when it uses ****Recovery Point Objective**** with this exact governed meaning: The maximum tolerable loss or unavailability of recoverable state measured from the disruption point for a declared domain.

Counterexample: A dependent artifact uses ****Recovery Point Objective**** in a way that violates its required distinction: RPO is a target, not proof that recovery met it.

Relationships: None registered

Lifecycle: None registered

Constraint: RPO is a target, not proof that recovery met it.

HAL-TERM-0148 — Recovery Time Objective

The target duration for restoring a declared service or capability to its required state after disruption.

Distinction: RTO does not authorize unsafe shortcuts during recovery.

Type/category/status: Recovery target / Operations / Approved

Aliases: RTO

Source basis: Direct source normalization

Book I: Book I Decisions 38, 42, and 50

Book II: Book II Chapters 27 and 28

Book III: Book III Chapters 5, 6, 7, and 8

Example: A dependent artifact cites `HAL-TERM-0148` when it uses ****Recovery Time Objective**** with this exact governed meaning: The target duration for restoring a declared service or capability to its required state after disruption.
Counterexample: A dependent artifact uses ****Recovery Time Objective**** in a way that violates its required distinction: RTO does not authorize unsafe shortcuts during recovery.

Relationships: None registered

Lifecycle: None registered

Constraint: RTO does not authorize unsafe shortcuts during recovery.

HAL-TERM-0149 — Release

A versioned, traceable, qualified set of software, configuration, schemas, models, and deployment artifacts approved for a declared environment and scope.

Distinction: A successful build is not a Release.

Type/category/status: Governed artifact set / Engineering / Approved

Aliases: None

Source basis: Engineering term normalized under Books I–III

Book I: Book I Decisions 43, 50, and 58

Book II: Book II Chapters 29 and 35

Book III: Book III Chapters 1, 7, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0149` when it uses ****Release**** with this exact governed meaning: A versioned, traceable, qualified set of software, configuration, schemas, models, and deployment artifacts approved for a declared environment and scope.

Counterexample: A dependent artifact uses ****Release**** in a way that violates its required distinction: A successful build is not a Release.

Relationships: HAL-REL-0054

Lifecycle: HAL-TRANS-0025, HAL-TRANS-0026

Constraint: A successful build is not a Release.

HAL-TERM-0150 — Migration

A governed transition of data, state, schema, interface, configuration, or runtime behavior from one compatible condition to another.

Distinction: A Migration must distinguish reversible steps from irreversible effects and compensation.

Type/category/status: State or contract change / Engineering / Approved

Aliases: None

Source basis: Engineering term normalized under Books I–III

Book I: Book I Decisions 43, 50, and 58

Book II: Book II Chapters 29 and 35

Book III: Book III Chapters 1, 7, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0150` when it uses ****Migration**** with this exact governed meaning: A governed transition of data, state, schema, interface, configuration, or runtime behavior from one compatible condition to another.

Counterexample: A dependent artifact uses ****Migration**** in a way that violates its required distinction: A Migration must distinguish reversible steps from irreversible effects and compensation.

Relationships: None registered

Lifecycle: None registered

Constraint: A Migration must distinguish reversible steps from irreversible effects and compensation.

HAL-TERM-0151 — Architecture Decision Record

A durable record of a consequential architecture decision, context, alternatives, rationale, consequences, source traceability, and review status.

Distinction: An ADR cannot authorize deviation from Book II without the approved architecture-governance process.

Type/category/status: Decision record / Engineering / Approved

Aliases: ADR

Source basis: Engineering term normalized under Books I–III

Book I: Book I Decisions 43, 50, and 58

Book II: Book II Chapters 29 and 35

Book III: Book III Chapters 1, 7, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0151` when it uses ****Architecture Decision Record**** with this exact governed meaning: A durable record of a consequential architecture decision, context, alternatives, rationale, consequences, source traceability, and review status.

Counterexample: A dependent artifact uses ****Architecture Decision Record**** in a way that violates its required distinction: An ADR cannot authorize deviation from Book II without the approved architecture-governance process.

Relationships: HAL-REL-0047

Lifecycle: None registered

Constraint: An ADR cannot authorize deviation from Book II without the approved architecture-governance process.

HAL-TERM-0152 — Architecture Deviation

A documented departure from an applicable Book II requirement processed through architecture governance with scope, risk, evidence, approval, and expiry or remediation.

Distinction: It is not an ordinary Book III control exception and cannot amend Book II silently.

Type/category/status: Governed exception class / Engineering / Approved

Aliases: None

Source basis: Engineering term normalized under Books I–III

Book I: Book I Decisions 43, 50, and 58

Book II: Book II Chapters 29 and 35

Book III: Book III Chapters 1, 7, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0152` when it uses **Architecture Deviation** with this exact governed meaning: A documented departure from an applicable Book II requirement processed through architecture governance with scope, risk, evidence, approval, and expiry or remediation.

Counterexample: A dependent artifact uses **Architecture Deviation** in a way that violates its required distinction: It is not an ordinary Book III control exception and cannot amend Book II silently.

Relationships: None registered

Lifecycle: None registered

Constraint: It is not an ordinary Book III control exception and cannot amend Book II silently.

HAL-TERM-0153 — Control

A stable, attributable requirement with applicability, responsibility, enforcement, evidence, severity, exception authority, verification, and source traceability.

Distinction: Advice is not a Control unless it is made objectively reviewable.

Type/category/status: Enforceable rule record / Engineering / Approved

Aliases: None

Source basis: Engineering term normalized under Books I–III

Book I: Book I Decisions 43, 50, and 58

Book II: Book II Chapters 29 and 35

Book III: Book III Chapters 1, 7, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0153` when it uses **Control** with this exact governed meaning: A stable, attributable requirement with applicability, responsibility, enforcement, evidence, severity, exception authority, verification, and source traceability.

Counterexample: A dependent artifact uses **Control** in a way that violates its required distinction: Advice is not a Control unless it is made objectively reviewable.

Relationships: HAL-REL-0048

Lifecycle: None registered

Constraint: Advice is not a Control unless it is made objectively reviewable.

HAL-TERM-0154 — Exception

A documented, scoped, risk-assessed, compensating, approved, expiring departure from a waivable lower-order Control.

Distinction: A constitutional invariant cannot be waived; an Architecture Deviation uses its own governance path.

Type/category/status: Time-bounded control relief / Engineering / Approved

Aliases: waiver

Source basis: Engineering term normalized under Books I–III

Book I: Book I Decisions 43, 50, and 58

Book II: Book II Chapters 29 and 35

Book III: Book III Chapters 1, 7, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0154` when it uses **Exception** with this exact governed meaning: A documented, scoped, risk-assessed, compensating, approved, expiring departure from a waivable lower-order Control.

Counterexample: A dependent artifact uses **Exception** in a way that violates its required distinction: A constitutional invariant cannot be waived; an Architecture Deviation uses its own governance path.

Relationships: HAL-REL-0048

Lifecycle: HAL-TRANS-0027, HAL-TRANS-0028

Constraint: A constitutional invariant cannot be waived; an Architecture Deviation uses its own governance path.

HAL-TERM-0155 — Definition of Ready

The minimum evidenced conditions required before consequential implementation work may begin or enter its next

controlled stage.

Distinction: Readiness does not imply approval to release or cross the Reality Boundary.

Type/category/status: Entry criteria / Engineering / Approved

Aliases: DoR

Source basis: Engineering term normalized under Books I–III

Book I: Book I Decisions 43, 50, and 58

Book II: Book II Chapters 29 and 35

Book III: Book III Chapters 1, 7, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0155` when it uses **Definition of Ready** with this exact governed meaning: The minimum evidenced conditions required before consequential implementation work may begin or enter its next controlled stage.

Counterexample: A dependent artifact uses **Definition of Ready** in a way that violates its required distinction: Readiness does not imply approval to release or cross the Reality Boundary.

Relationships: None registered

Lifecycle: None registered

Constraint: Readiness does not imply approval to release or cross the Reality Boundary.

HAL-TERM-0156 — Definition of Done

The minimum evidenced conditions required before work may be treated as complete within a declared scope.

Distinction: Done does not erase post-release monitoring, retention, or recovery obligations.

Type/category/status: Completion criteria / Engineering / Approved

Aliases: DoD

Source basis: Engineering term normalized under Books I–III

Book I: Book I Decisions 43, 50, and 58

Book II: Book II Chapters 29 and 35

Book III: Book III Chapters 1, 7, 8, and 9

Example: A dependent artifact cites `HAL-TERM-0156` when it uses **Definition of Done** with this exact governed meaning: The minimum evidenced conditions required before work may be treated as complete within a declared scope.

Counterexample: A dependent artifact uses **Definition of Done** in a way that violates its required distinction: Done does not erase post-release monitoring, retention, or recovery obligations.

Relationships: None registered

Lifecycle: None registered

Constraint: Done does not erase post-release monitoring, retention, or recovery obligations.

HAL-TERM-0166 — Release Authority

The Book III-designated role authorized to certify a qualified Release for a declared deployment scope after all required verification and reviews have passed.

Distinction: Release Authority cannot waive Constitutional Invariants, replace architecture, security, or privacy review, or authorize deployment beyond the certified scope.

Type/category/status: Governed certification role / Engineering / Approved

Aliases: None

Source basis: Engineering term normalized under Books I–III

Book I: Book I Decisions 43, 50, and 58

Book II: Book II Chapters 29 and 35

Book III: Book III Chapter 7

Example: The Release Authority certifies Release `R-42` only after the scoped qualification evidence and required architecture, security, and privacy reviews pass.

Counterexample: A successful CI build deploys itself because pipeline success is treated as Release Authority approval.

Relationships: HAL-REL-0054

Lifecycle: None registered

Constraint: Release Authority cannot waive Constitutional Invariants, replace architecture, security, or privacy review, or authorize deployment beyond the certified scope.

Relationship and lifecycle rules

Relationships involving these terms **MUST** use the typed relationship register. Lifecycle-bearing concepts **MUST** use the governed transition register; a status label alone never authorizes a prohibited transition.

Term-specific semantic evidence

Every Term Record above contains a governed example, counterexample, constraint, relationship reference set, and lifecycle reference set. “None registered” is explicit.

Anti-patterns

Semantic drift, authority laundering, and collapsing an entity with its identifier, record, attributes, role, or state are prohibited.

Verification

Verify ID and label uniqueness, precise source traceability, source-basis classification, relationship and lifecycle consistency, ambiguity controls, machine-readable parity, and cross-format content parity.

Change and deprecation

Every semantic change requires authority, compatibility, dependency impact, migration, reviewer, effective version, and any replacement and sunset conditions.

Review findings

Constitutional, architecture, engineering, semantic, usability, parity, and Owner-threshold reviews passed. No Owner Review item.